

QUESTIONS 30 / 48 completed

Arithmetic

ARITHMETIC

Write a program to read in two integers and perform the following operations on them: addition, subtraction, multiplication, division, and modulo.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a=6,b=3;
5     cout<<"the addition of a and b is:"<<(a+b)<<endl;
6     cout<<"the subtraction of a and b is:"<<(a-b)<<endl;
7     cout<<"the multiplication of a and b is:"<<(a*b)<<endl;
8     cout<<"the division of a and b is:"<<(a/b)<<endl;
9     cout<<"the modulo of a and b is:"<<(a%b)<<endl;
10    return 0;
11 }
```

OUTPUT

```
the addition of a and b is:9
the subtraction of a and b is:3
the multiplication of a and b is:18
the division of a and b is:2
the modulo of a and b is:0
```

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Average

AVERAGE

Write a program to read in three integers and compute the average of the three numbers. Use the assignment operator to store the result in a variable.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int a=2,b=2,c=2;
6     int avg=(a+b+c)/3;
7     cout<<"the average of three numbers is:"<<avg;
8     return 0;
9 }
```

OUTPUT

```
the average of three numbers is:2
```

QUESTIONS 30 / 48 completed

Compare ▾

COMPARE

Write a program to read in two integers and determine if they are equal or not.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a=12,b=12;
5     if(a==b)cout<<"equal";
6     else cout<<"not equal";
7     return 0;
8 }
```

OUTPUT

```
equal
```

QUESTIONS 30 / 48 completed

Floating-Point ▾

FLOATING-POINT

Write a program to read in two floating-point numbers and perform the following operations on them: addition, subtraction, multiplication, and division.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     float a=5,b=3;
5     cout<<"addition:"<<a+b<<endl;
6     cout<<"subtraction:"<<a-b<<endl;
7     cout<<"multiplication:"<<a*b<<endl;
8     if(b!=0)
9         cout<<"division:"<<a/b<<endl;
10    else
11        cout<<"division:cannot divide by zero!"<<endl;
12    return 0;
13
14 }
```

OUTPUT

```
addition:8
subtraction:2
multiplication:15
division:1.66667
```

QUESTIONS 30 / 48 completed

Even-Odd

EVEN-ODD

Write a program to read in an integer and determine if it is even or odd.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n=7;
5     if(n%2==0){
6         cout<<n<<"is even"<<endl;
7     }
8     else{
9         cout<<n<<"is odd"<<endl;
10    }
11    return 0;
12 }
```

OUTPUT

7is odd

QUESTIONS 30 / 48 completed

Pos-Neg

POS-NEG

Write a program to read in an integer and determine if it is positive, negative, or zero.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int n=6;
6     if(n>0) cout<<"positive";
7     else if(n<0)cout<<"negative";
8     else cout<<"zero";
9     return 0;
10 }
```

OUTPUT

positive

QUESTIONS 30 / 48 completed

Vow-Cons

VOW-CONS

Write a program to read in a character and determine if it is a vowel or a consonant.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     char ch;
5     cin>>ch;
6     if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' || ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
7         cout<<"vowel";
8     else
9         cout<<"consonant";
10    return 0;
11 }
```

OUTPUT

vowel

QUESTIONS 30 / 48 completed

Greater

GREATER

Write a program to read in two integers and use the conditional operator to determine which number is greater.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a=3,b=5;
5     int greater=(a>b)?a:b;
6     cout<<greater;
7     return 0;
8 }
```

OUTPUT

5

QUESTIONS 30 / 48 completed

Swap

SWAP

Write a program to read in two integers and swap their values using the bitwise XOR operator.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int a=3,b=4;
6     if(a!=b){
7         a=a^b;
8         b=a^b;
9         a=a^b;
10    }
11    }
12    cout<<"a="<<a<<"",b="<<b;
13    return 0;
14
15
16 }
```

OUTPUT

a=4, b=3

QUESTIONS 30 / 48 completed

Nearest

NEAREST

Write a program to read in a floating-point number and round it to the nearest integer using the floor and ceil functions.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 #include<cmath>
3 using namespace std;
4 int main(){
5     double x=2.7;
6     double f=floor(x);
7     double c=ceil(x);
8     if(x-f<c-x)
9         cout<<(int)f;
10    else
11        cout<<(int)c;
12    return 0;
13 }
14
```

OUTPUT

3